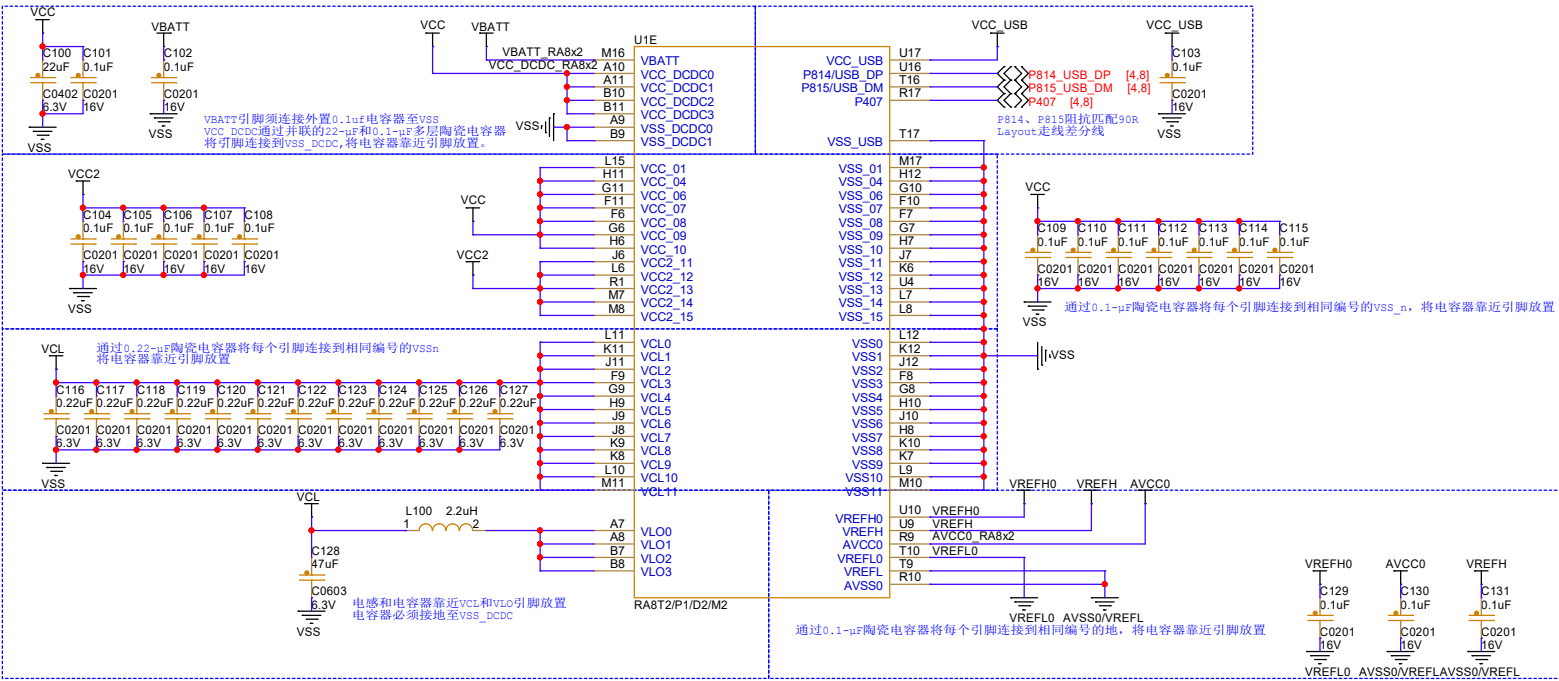




www.gitee.com/
ramcu/
cpk_examples

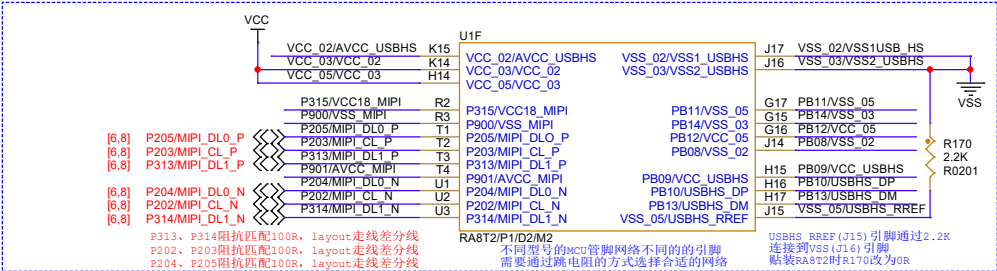


www.github.com/
renesas/
cpk_examples



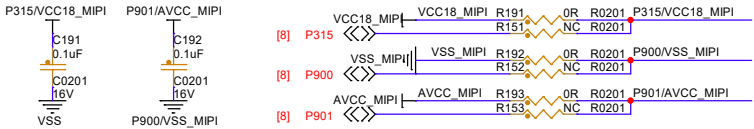
核心板省功能阻容设置

核心板芯片	R7KA8T2L	R7KA8P1K
C168-169	0.1uF	NC
C170	2.2uF	2.2uF
C171-C174	NC	0.1uF
C191-C192	NC	0.1uF
R145-R150	NC	NC
R151-R168	0R	NC
R170	0R	2.20K 1%
L171	NC	120R 100MHz
R172-R180	NC	0R
R191-R193	NC	0R



MIPI部分

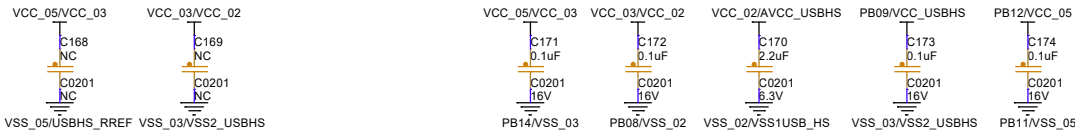
连接电源的器件靠近MCU摆放，连接信号的器件在MCU-BTB间任意摆放



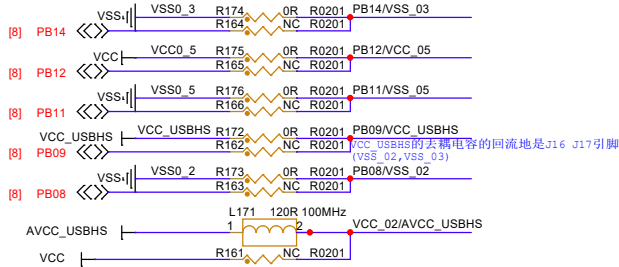
电阻选装

由于RA8P1/D2/M2和RA8T2的部分管脚网络不同，通过选装阻容的方式，应对不同型号的IC贴装。默认贴装RA8P1/D2/M2

连接电源的器件靠近MCU摆放，连接信号的器件在MCU-BTB间任意摆放



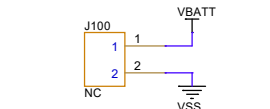
与电源相关的引脚，MCU信号出来后立即用电阻隔离



差分线走到J1 (BTB) 附近再用电阻隔离。



VBATT通孔接口



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Title CPKCOR-RA8P1核心板			
Size A3	Document Number MCU 电源	Rev V1.0	
Date: Saturday, September 12, 2026	Sheet 1	of 10	

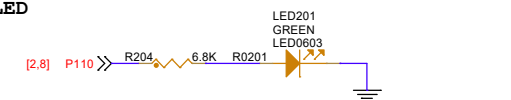
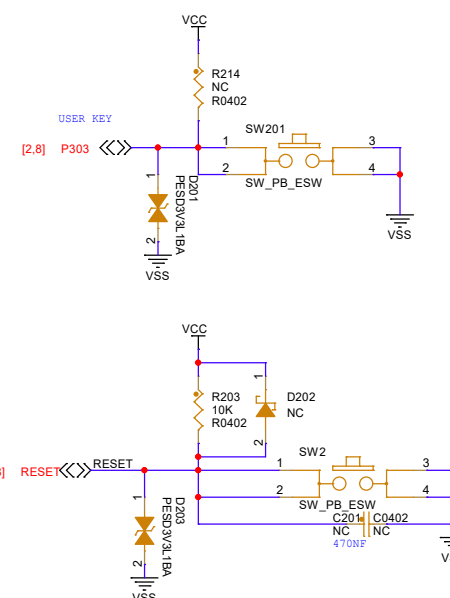


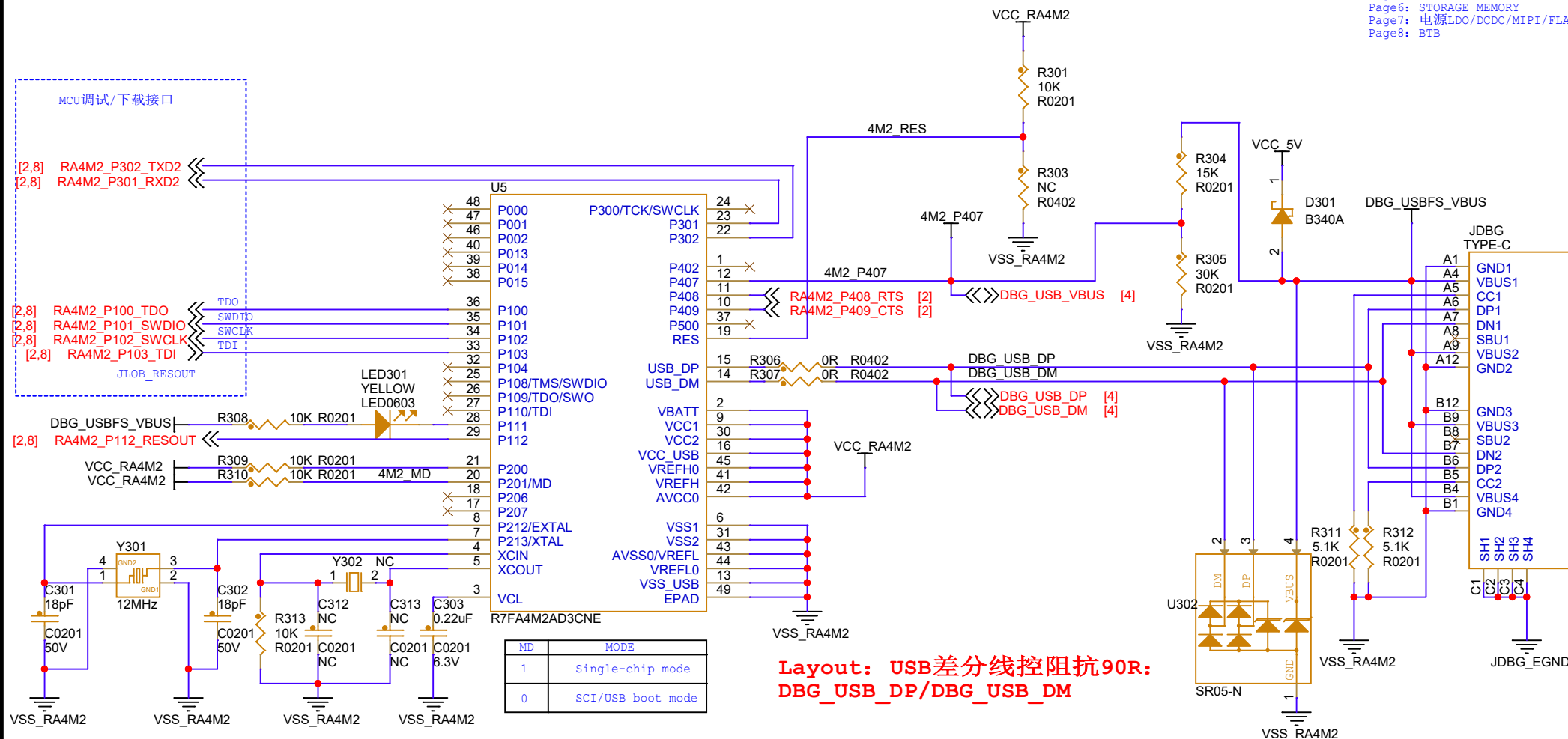
Figure 1: Pin connection diagram of the RA4M2 module. The diagram shows the RA4M2 module (TXB0108RGYR) connected to a microcontroller (MCU) and an external signal source (J-LINK). The RA4M2 module has pins for VCC, GND, and various signals. The MCU interface includes pins for TDI, TDO, PD03, PD02, PD04, PD05, RXD0, TXD0, RXD2, TXD2, TDI, TDO, and RTS. The external signal source (J-LINK) includes pins for RA4M2_P409_CTS, RA4M2_P301_RXD2, RA4M2_P100_TDO, RA4M2_P302_TDO, RA4M2_P103_TDI, RA4M2_P101_SWDIO, RA4M2_P102_SWCLK, and RA4M2_P408_RTS. The RA4M2 module also has pins for VCC, GND, and various signals like RXD0, TXD0, RXD2, TXD2, TDI, TDO, and RTS.

1.25mm间距测试点
顺序不要打乱

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Title CPKCOR-RA8P1核心板			
Size A3	Document Number MCU IO	Revision V1	
Date:	Saturday, September 12, 2026	Sheet 2	of 10

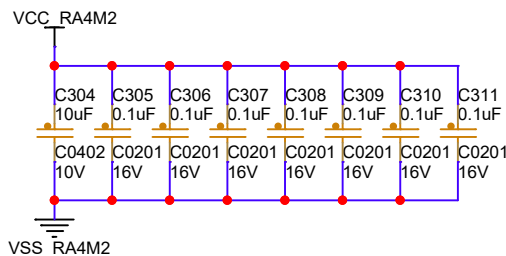
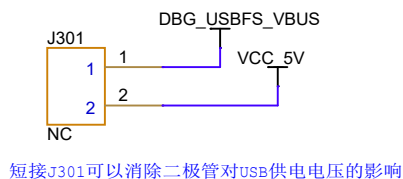
J-Link OB

Page1: MCU 电源
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Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB



Layout: USB差分线控阻抗90R:
DBG_USB_DP/DBG_USB_DM

注意: 只能使用12MHz晶振



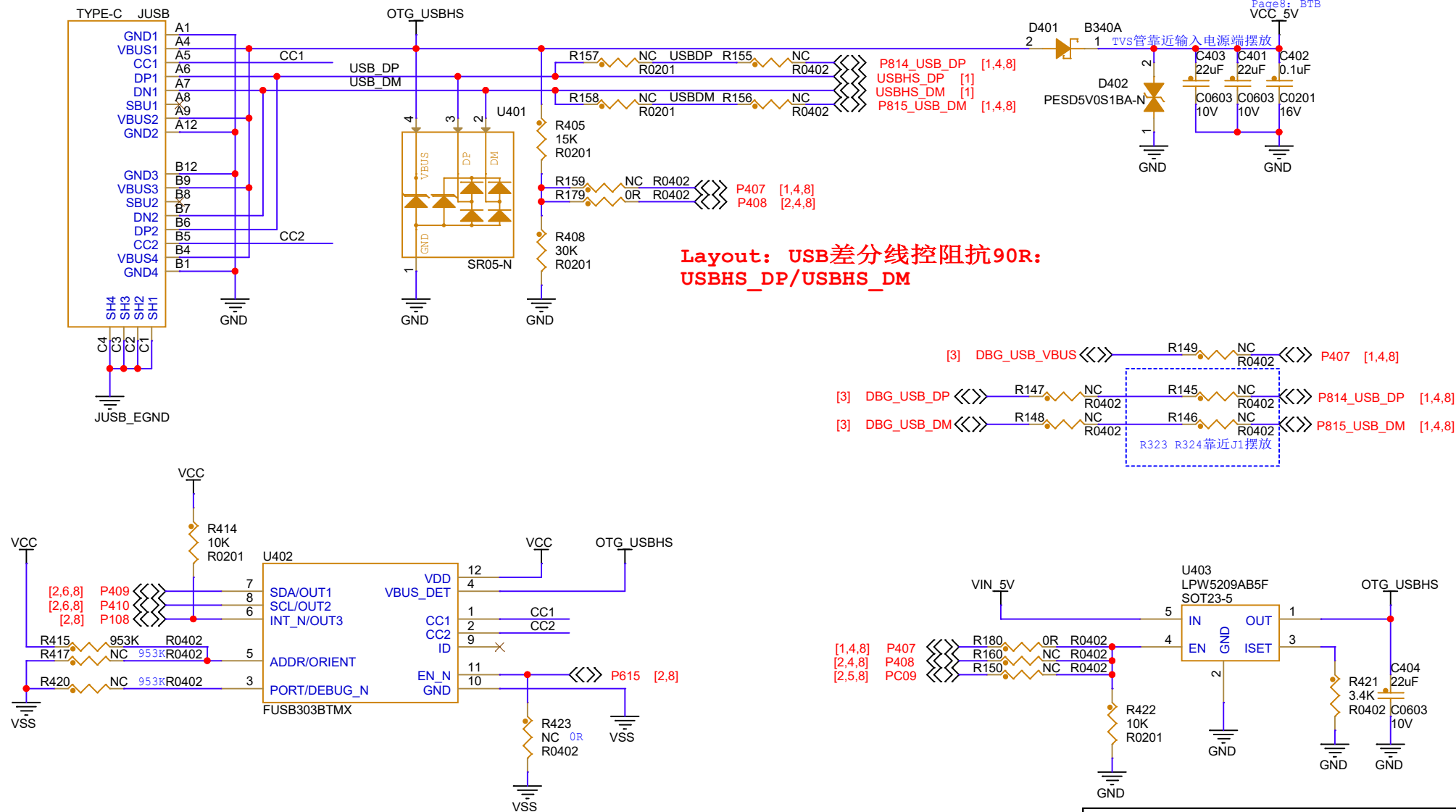
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Title CPKCOR-RA8P1核心板		
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USB HS

如果使用USB HS，主晶振只能选12/20/24/48MHz

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Page5: SDRAM
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Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB
VCC 5V



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Title

CPKCOR-RA8P1核心板

Size

A4

Document Number

USB HS

Rev

V1.0

Date:

Saturday, September 12, 2026

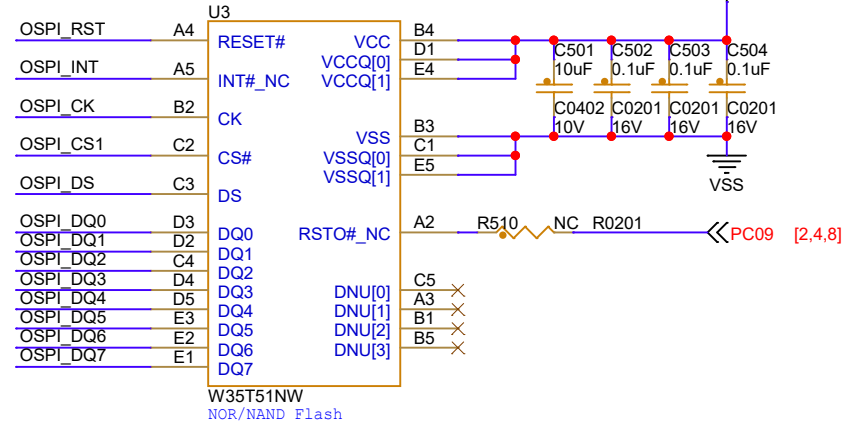
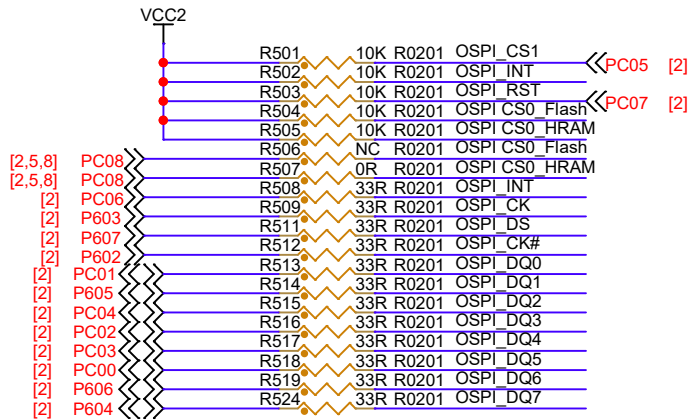
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of

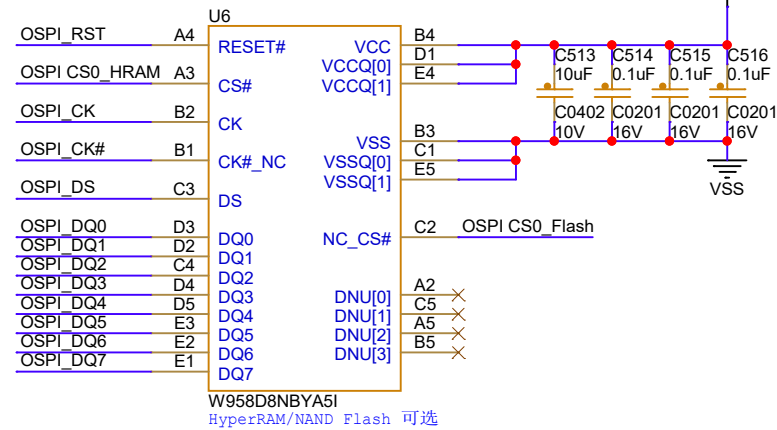
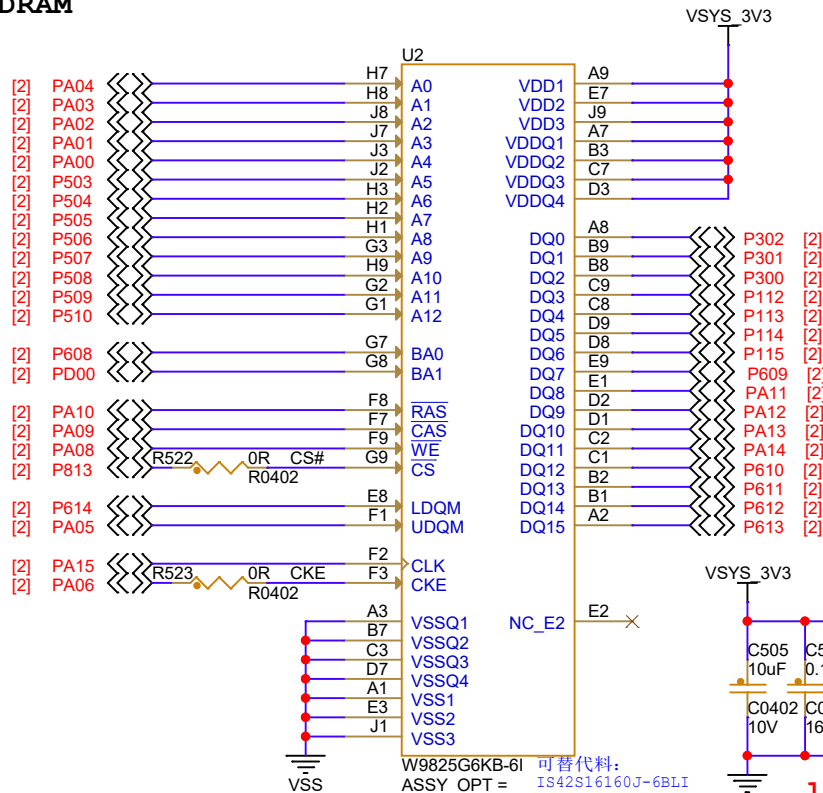
10

FLASH



Page1: MCU 电源
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Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB

SDRAM

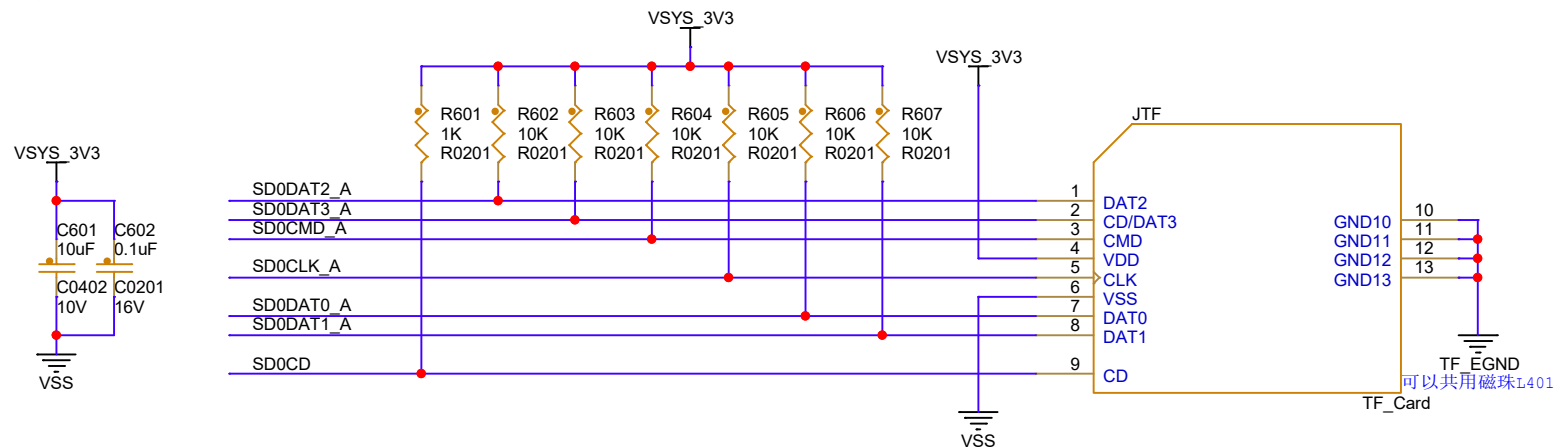


layout: SDRAM走线控阻抗50R

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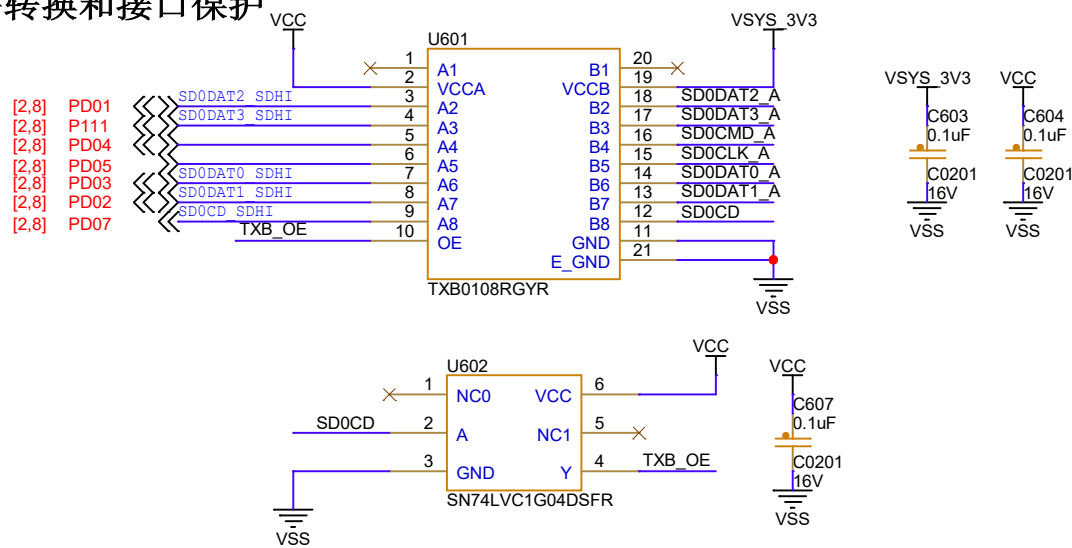
Title CPKCOR-RA8P1核心板		
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MicroSD卡槽

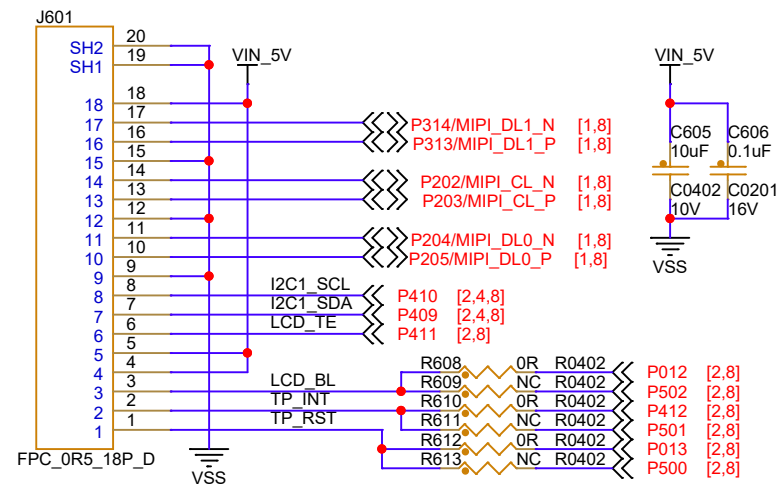


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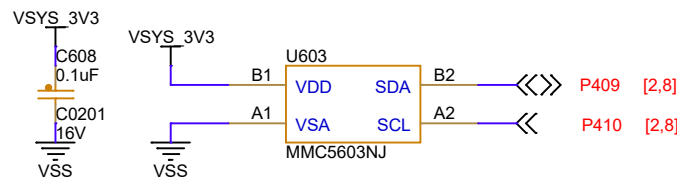
电平转换和接口保护



MIPI



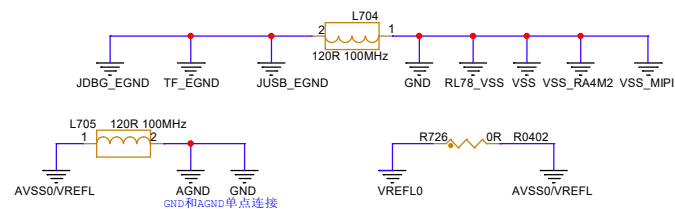
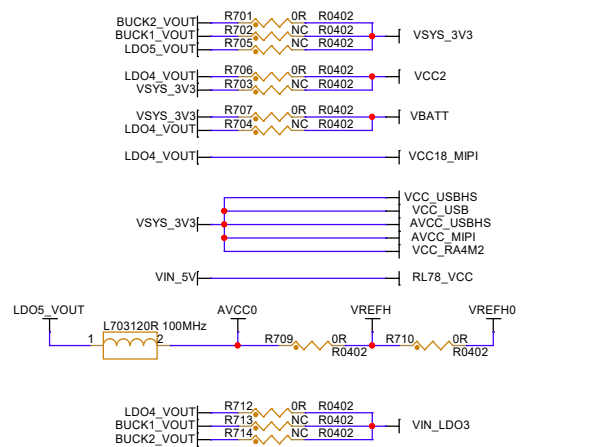
地磁传感器



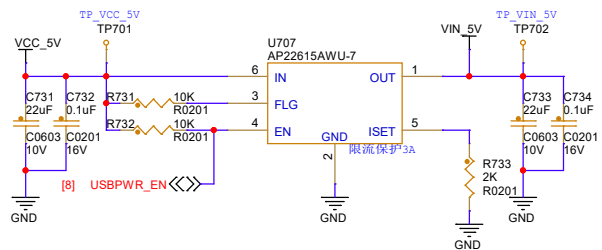
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Title CPKCOR-RA8P1核心板		
Size A4	Document Number SD CARD/MIPI	Rev V1.0
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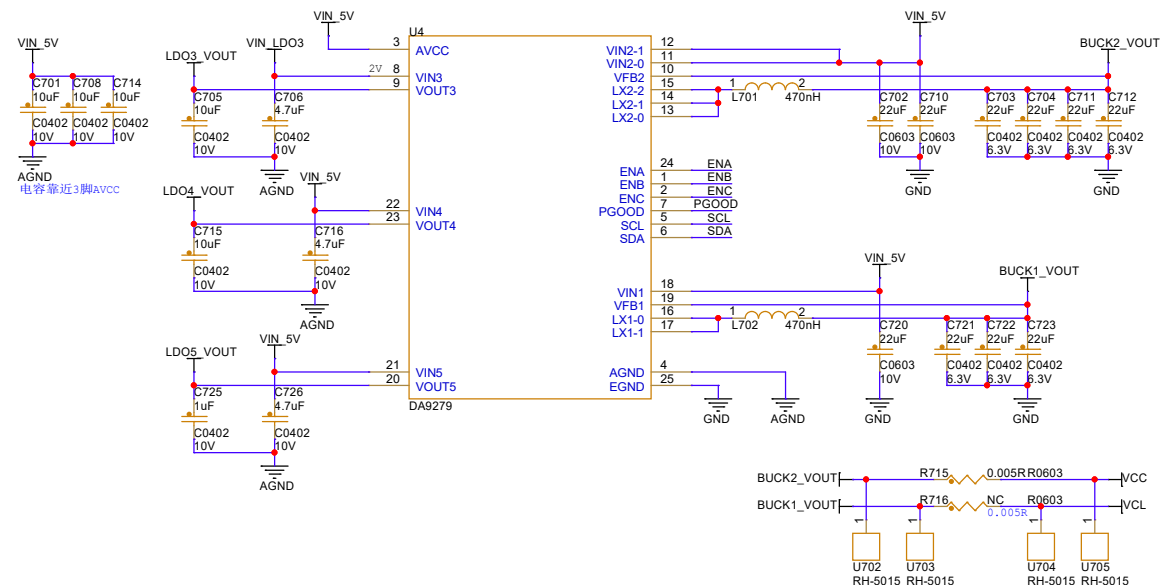
电源分割



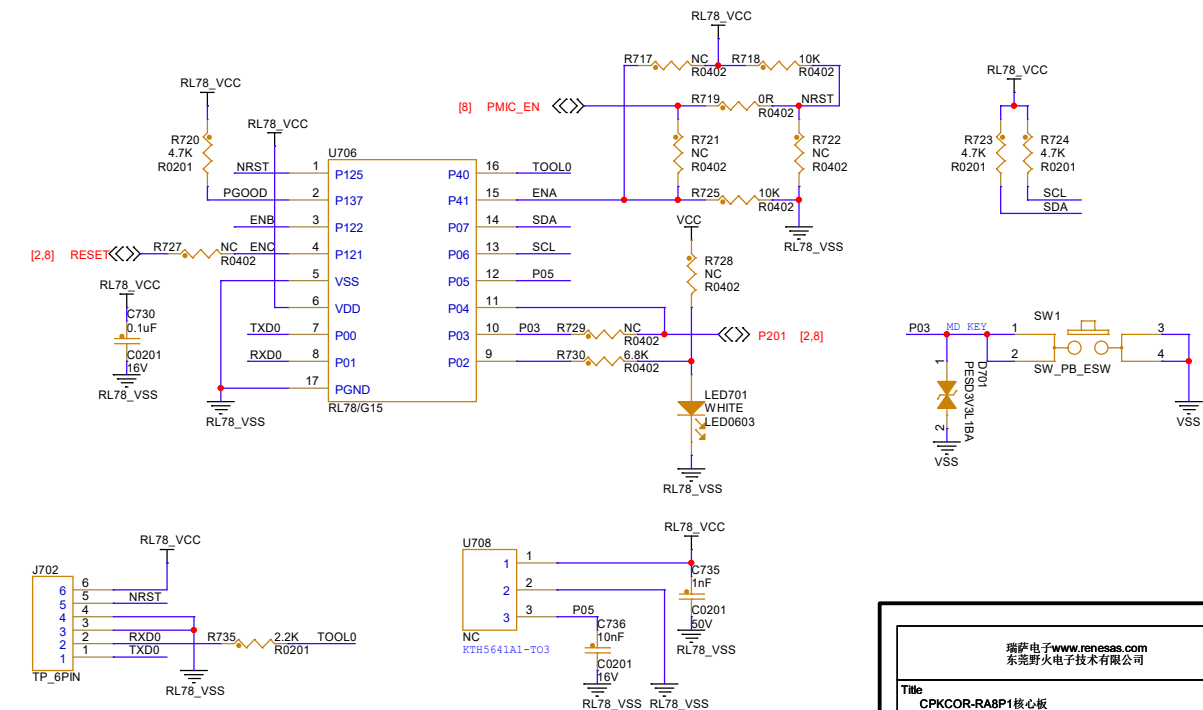
电源保护

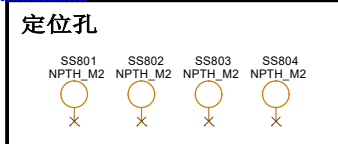
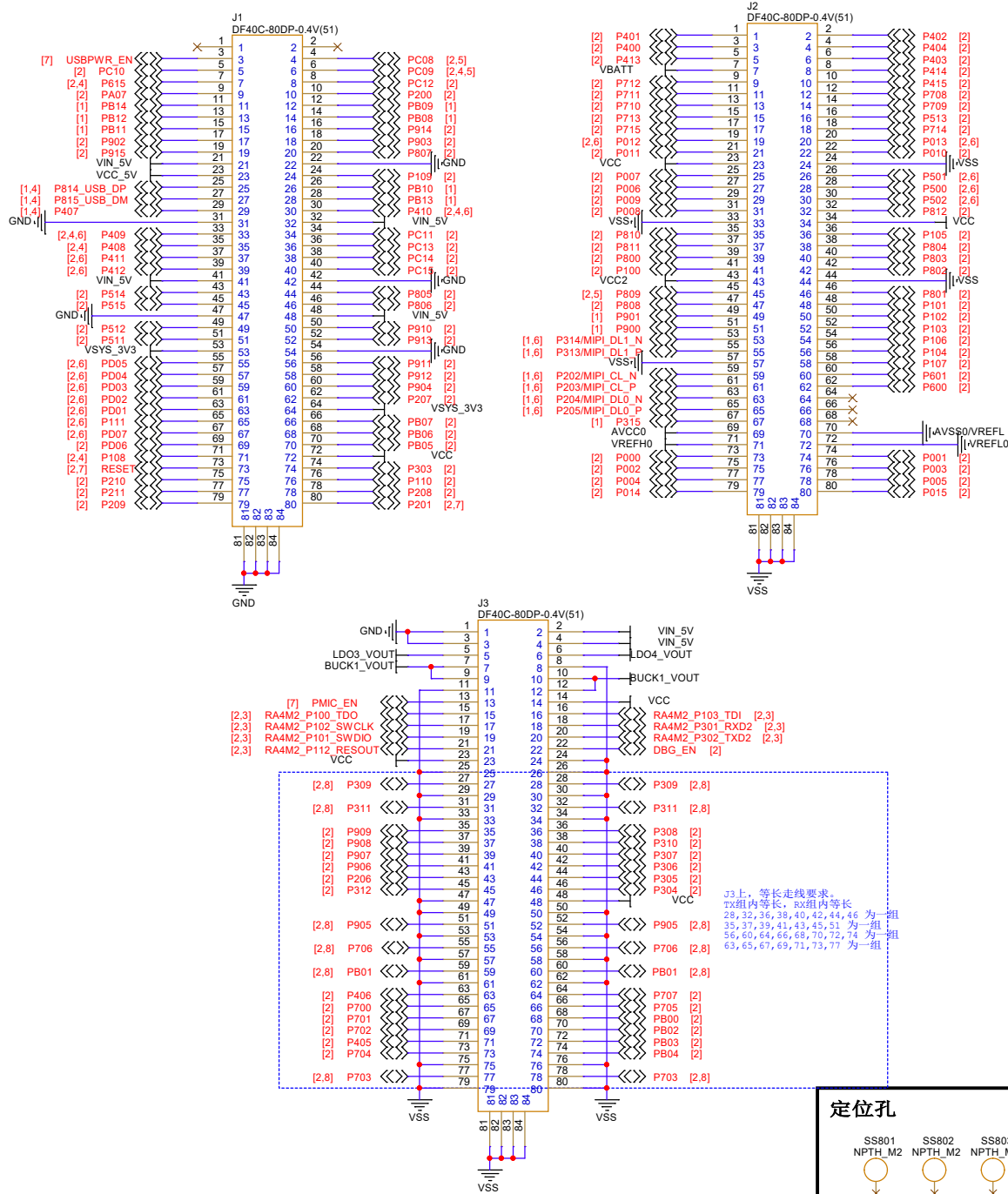


PMIC



MCU





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Title CPKCOR-RA8P1核心板		
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历史版本

版本号	日期	设计	描述
V1.0	2026-06-08	llm	初版发行

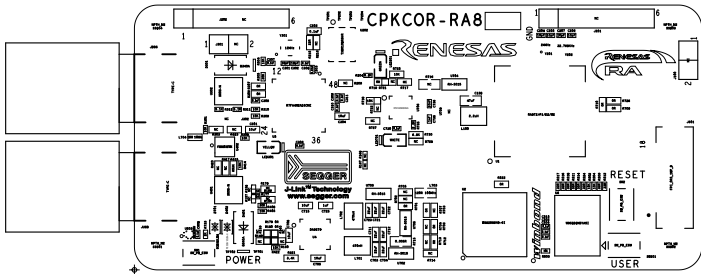
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CPKCOR-RA8P1核心板_原理图_V1.0

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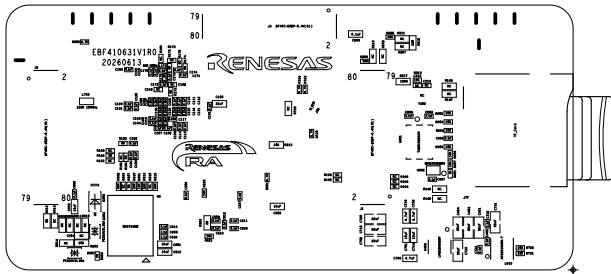
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ART FILM - ADT



ART FILM - ADT

ART FILM - ADB



ART FILM - ADB

CPKCOR-RA8P1 Power Supply Diagram

